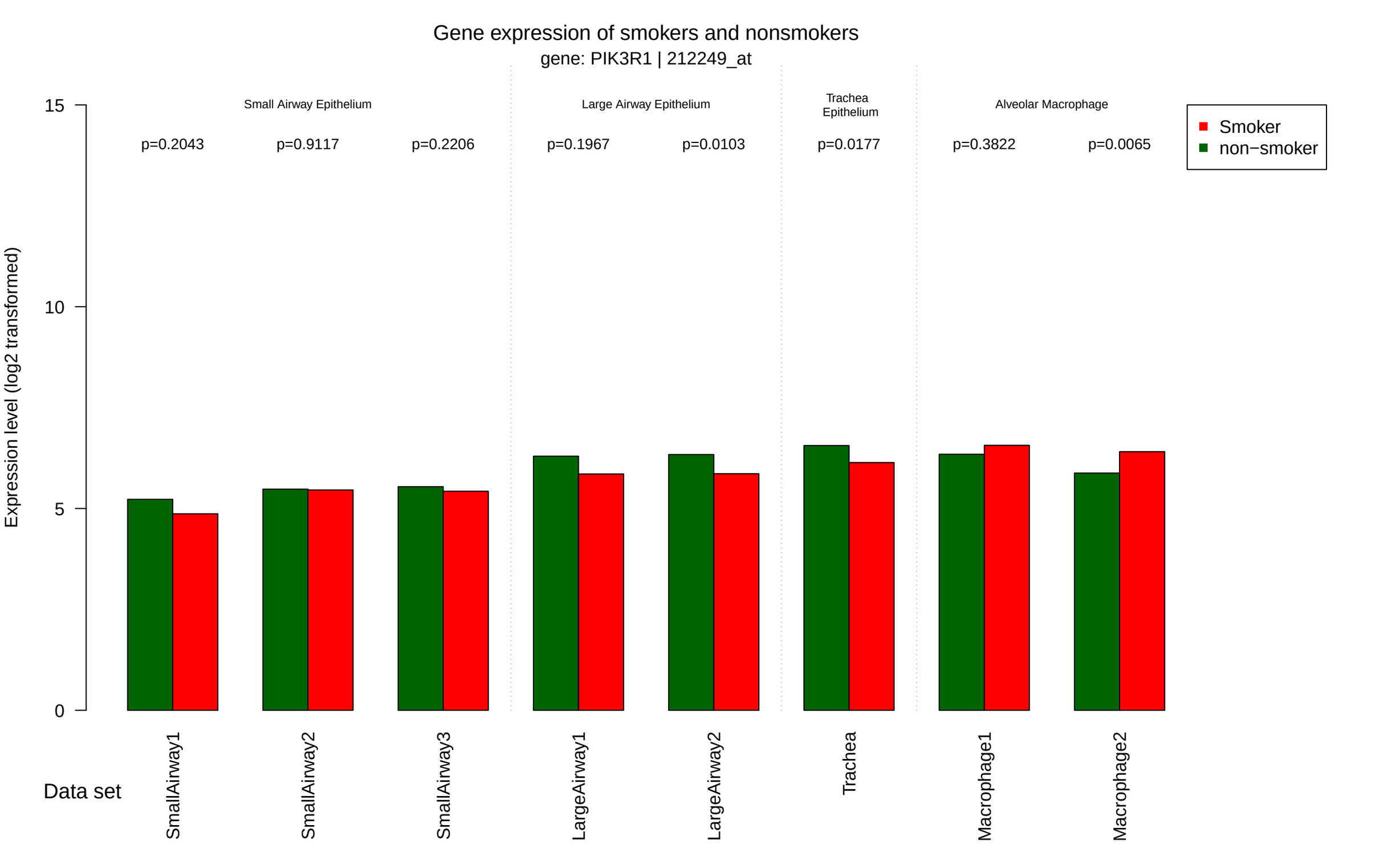
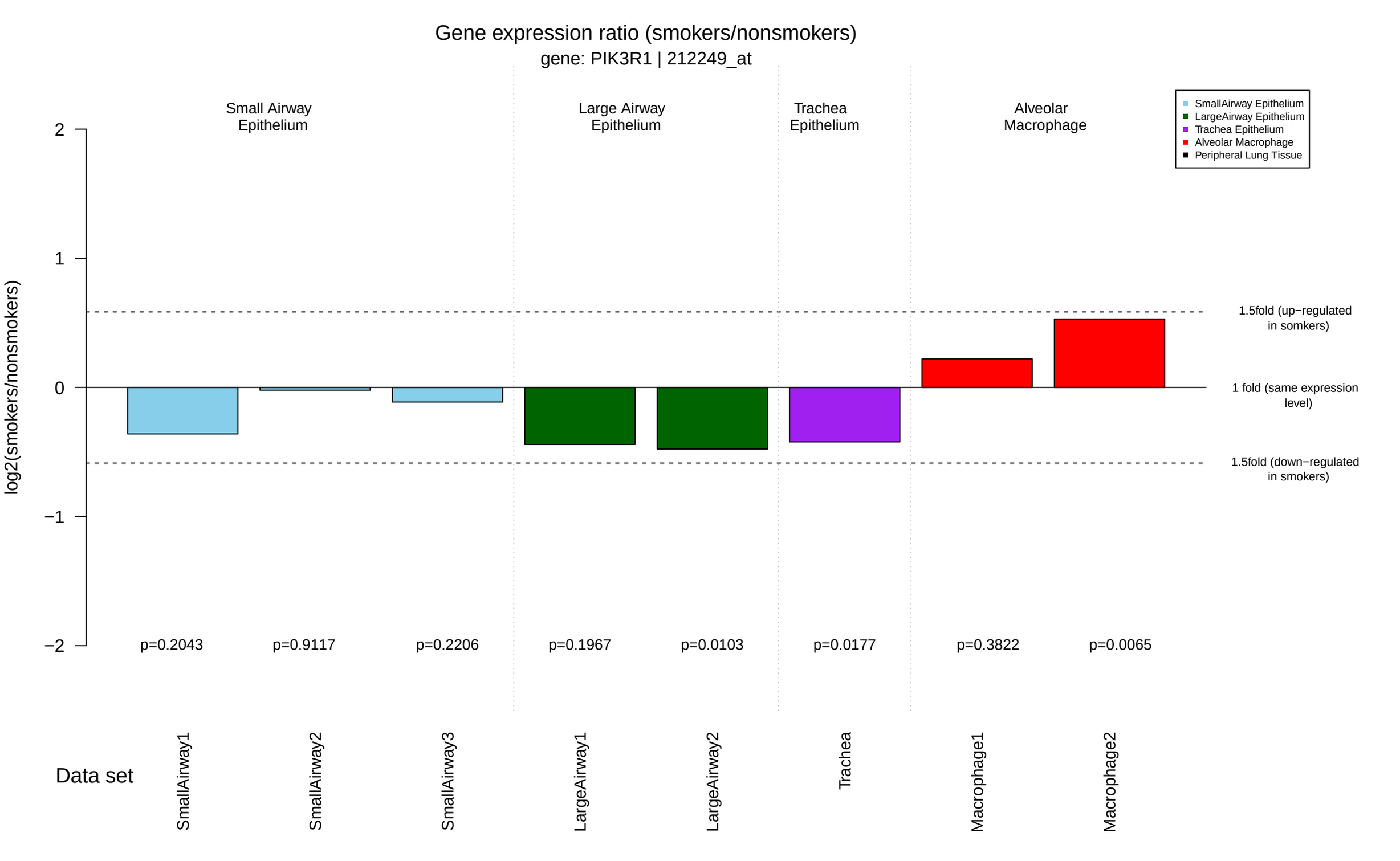
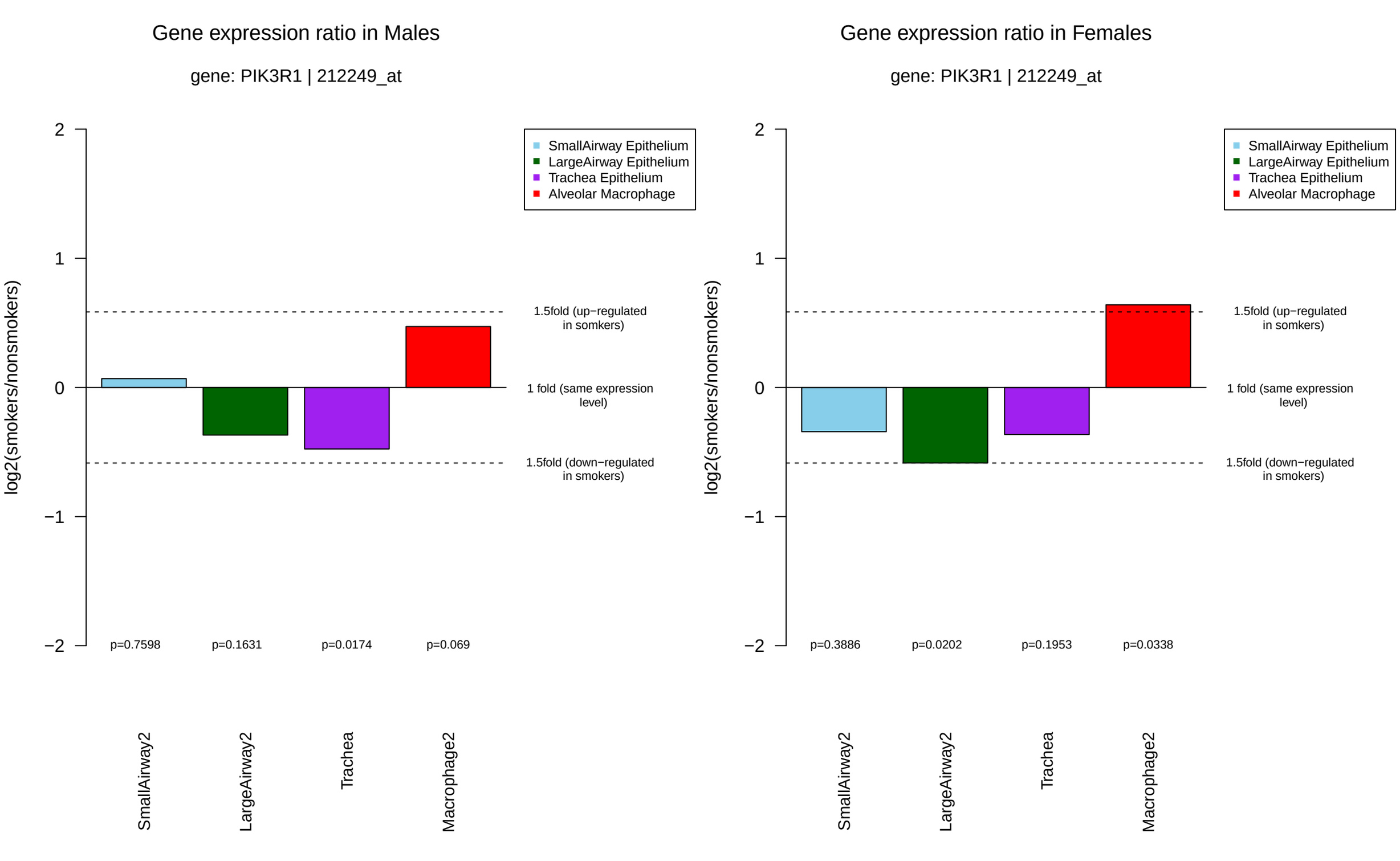
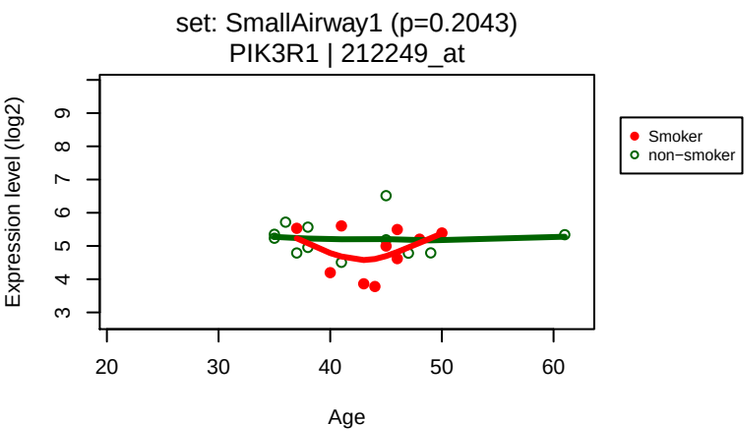
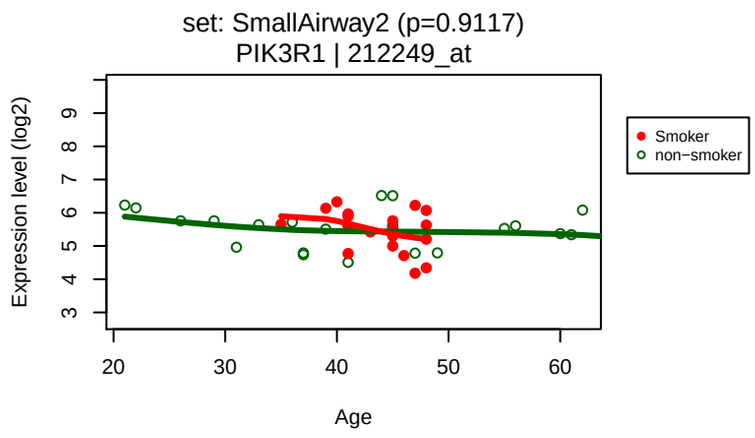
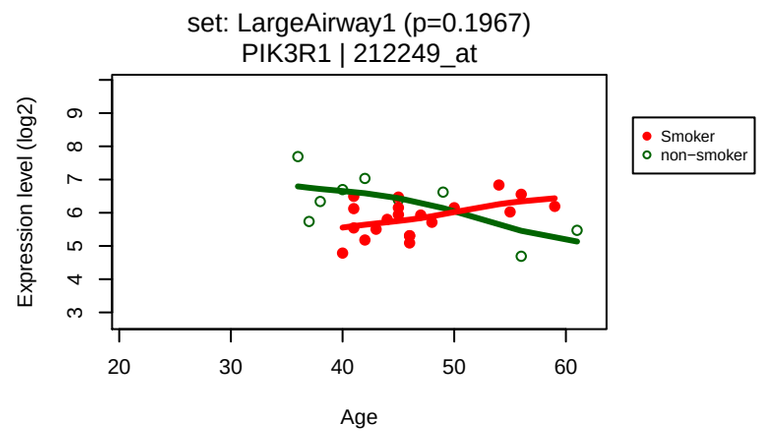

Gene expression of smokers and nonsmokers
gene: PIK3R1 | 212249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2043
p=0.9117
p=0.2206
p=0.1967
p=0.0103
p=0.0177
p=0.3822
p=0.0065
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

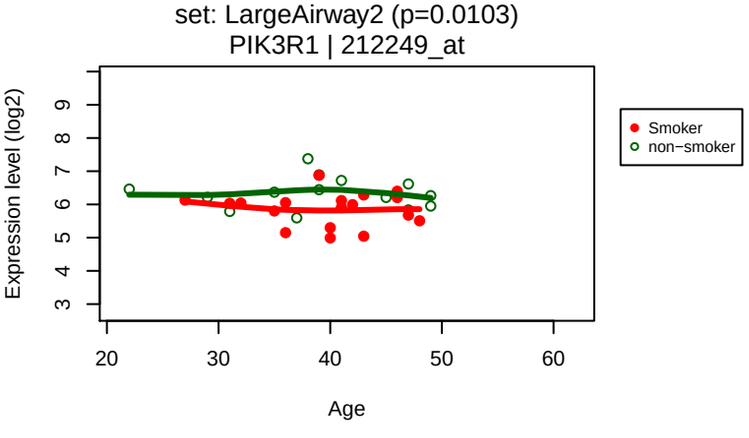

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R1 | 212249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2043
p=0.9117
p=0.2206
p=0.1967
p=0.0103
p=0.0177
p=0.3822
p=0.0065
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

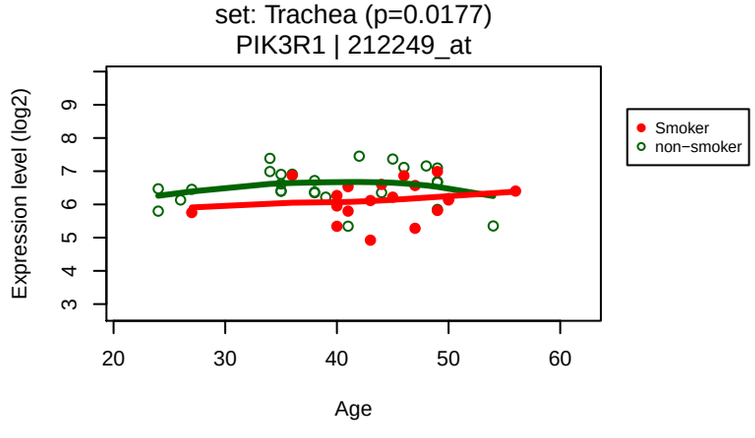

Gene expression ratio in Males
gene: PIK3R1 | 212249_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7598
p=0.1631
p=0.0174
p=0.069
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R1 | 212249_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3886
p=0.0202
p=0.1953
p=0.0338
SmallAirway2
LargeAirway2
Trachea
Macrophage2

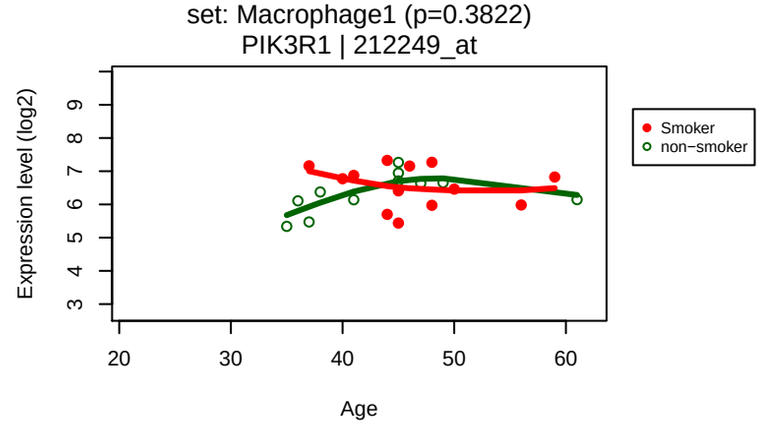

set: SmallAirway1 (p=0.2043)
PIK3R1 | 212249_at
Expression level (log2)
Age
Smoker
non−smoker

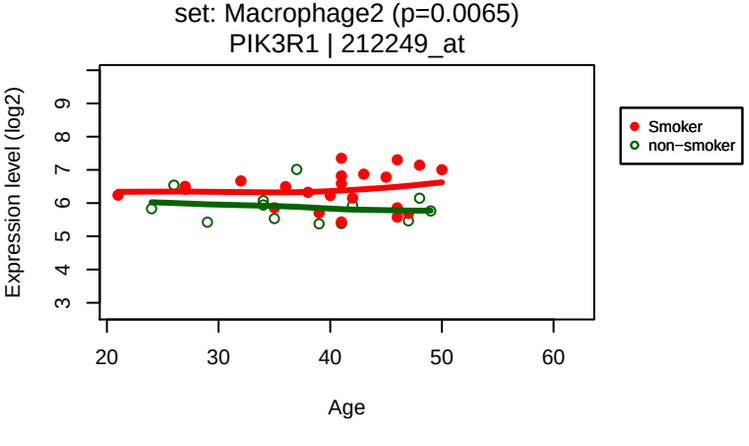

set: SmallAirway2 (p=0.9117)
PIK3R1 | 212249_at
Expression level (log2)
Age
Smoker
non−smoker

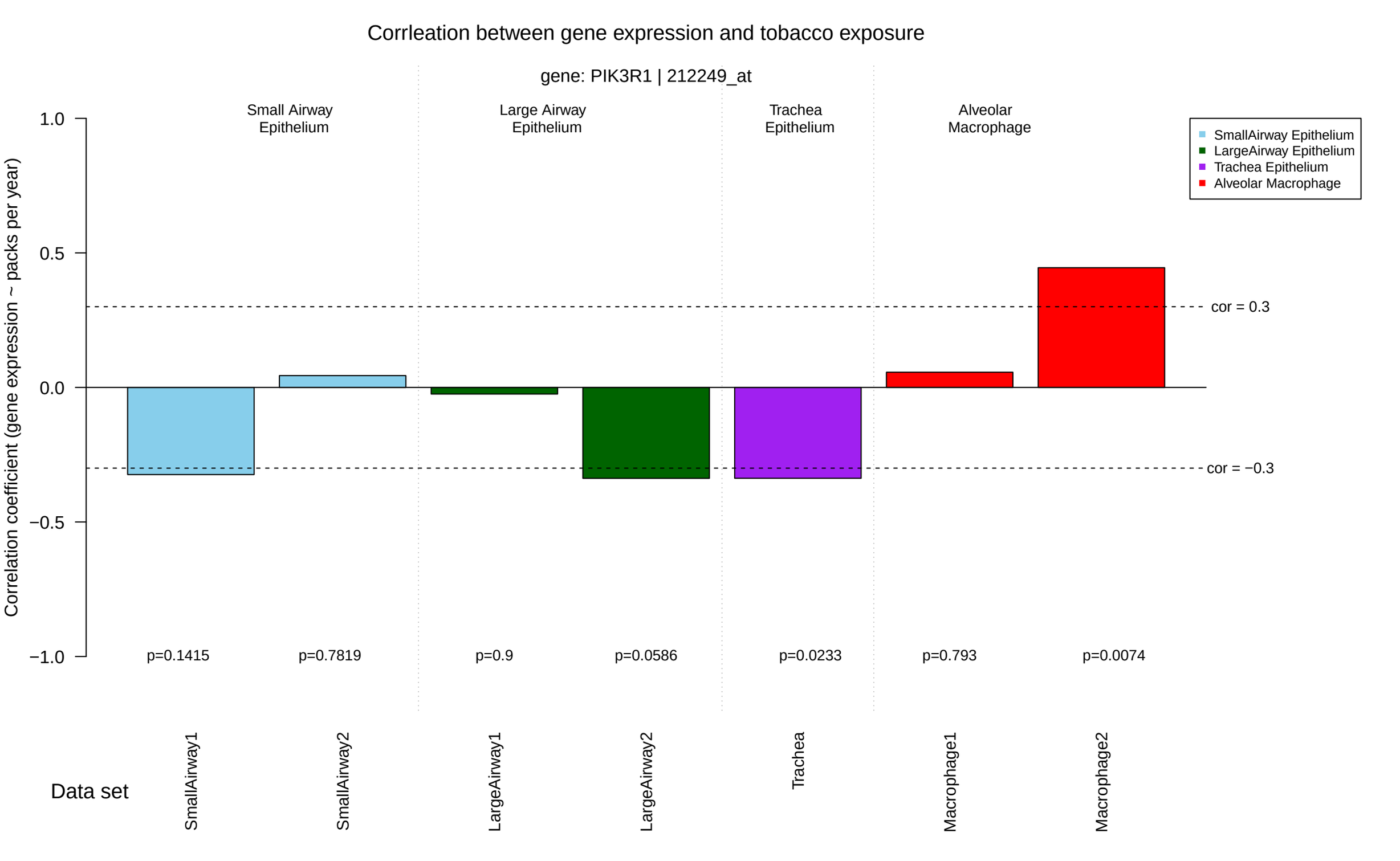

set: LargeAirway1 (p=0.1967)
PIK3R1 | 212249_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0103)
PIK3R1 | 212249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0177)
PIK3R1 | 212249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3822)
PIK3R1 | 212249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0065)
PIK3R1 | 212249_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R1 | 212249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1415
p=0.7819
p=0.9
p=0.0586
p=0.0233
p=0.793
p=0.0074
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2